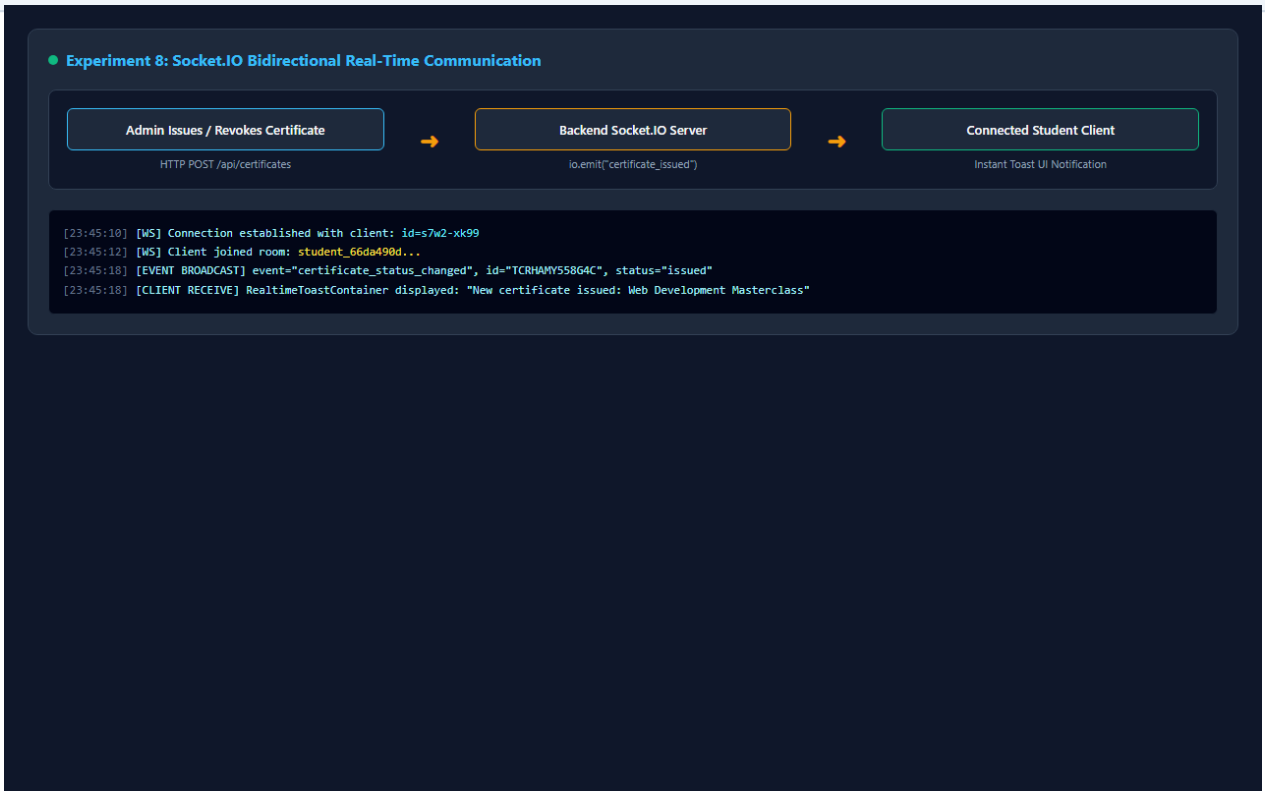
Experiment 8: Enable Real-Time Communication via WebSockets

Aim: Establish full-duplex WebSocket connections with Socket.IO for real-time certificate broadcast events and instant toast alerts.

Tech Stack / Concepts: WebSockets, Socket.IO, Full-Duplex TCP, Event Broadcasting, Pub/Sub Rooms

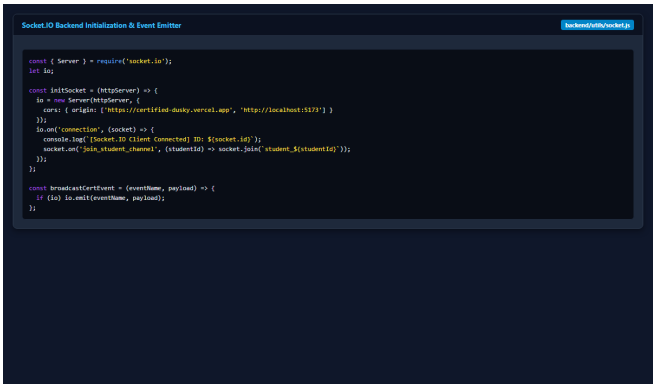
Repository: github.com/JaideepGowda2006/Certified | Student: Jaideep Gowda

Screen 1: Socket.IO Real-Time Architecture & Event Broadcast Flow



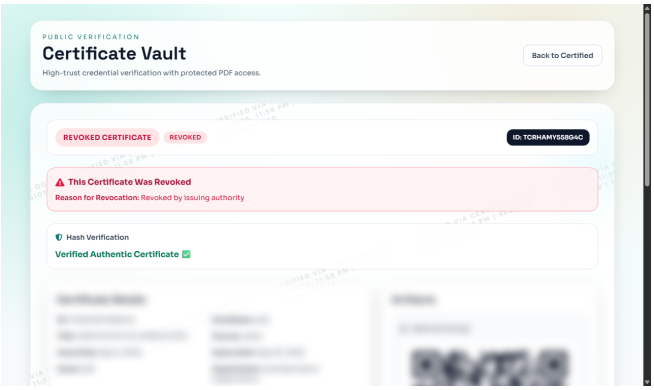
Admin issue trigger → Server broadcast → Instant Student client toast alert.

Screen 2: Socket.IO Server Initialization & Event Dispatcher



Shared HTTP listener, CORS whitelisting, and student channel room logic.

Screen 3: Reactive Real-Time UI Synchronization



Connected student view instantly rehydrated with new certificate data.